

Data Fairness Reflection

1. Basis

To judge whether a neighborhood is under-represented, I compared each neighborhood's share of library members with its share of checkouts. I used a difference of more than 5 percentage points below the neighborhood's member share as the threshold for clear under-representation. This provides a consistent basis for comparing the neighborhoods rather than judging them only by their total number of checkouts.

2. Finding

The comparison does not show a clearly under-represented neighborhood.

The five neighborhoods in the cleaned dataset were Heliopolis, Maadi, Nasr City, Shubra, and Zamalek. Shubra had 5 members, representing approximately 7.69% of the members, and 34 checkouts, representing approximately 8.42% of all checkouts. Heliopolis had 13 members and 87 checkouts, while Maadi had 20 members and 114 checkouts. Nasr City had 16 members and 101 checkouts, and Zamalek had 11 members and 68 checkouts.

The largest negative difference between member share and checkout share was very small compared with the 5-percentage-point threshold. Overall, the checkout shares are reasonably close to the neighborhoods' shares of members, so the data does not provide strong evidence that one neighborhood is under-represented.

3. A Plausible Reason

One possible reason for the small differences between neighborhoods is that members in different areas may use the library at slightly different rates. Factors such as distance from the library, availability of transportation, school or work schedules, and individual reading habits could affect how frequently members borrow books.

However, the current data does not contain enough information to identify the exact cause of these differences. The explanation should therefore be treated as a plausible possibility rather than a proven fact.

4. Next Step

Next summer, the library could monitor neighborhood membership and checkout activity again and compare the results with this year's baseline. If a larger gap appears in a future period, the library could investigate whether additional outreach, community events, or easier access to library services would help increase participation in the affected neighborhood.

Conclusion

Based on the cleaned dataset, no neighborhood appears clearly under-represented using the 5-percentage-point comparison rule. The member and checkout distributions are sufficiently similar that there is no strong evidence of a fairness problem in neighborhood representation.